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Student Number

Extension 2 Mathematics

HSC Assessment task 2

13th March 2015

General Instructions

- Reading time – 5 minutes
- Working time 3 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- In Questions 11-16 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks –

Section A

10 marks

- Attempt Questions 1 – 10
- Allow about 10 minutes for this section

Section B

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hour and 50 minutes for this section

Question	Marks
1-10	/10
11	/15
12	/15
13	/15
14	/15
15	/15
16	/15

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 10% of the Higher School Certificate Course Assessment

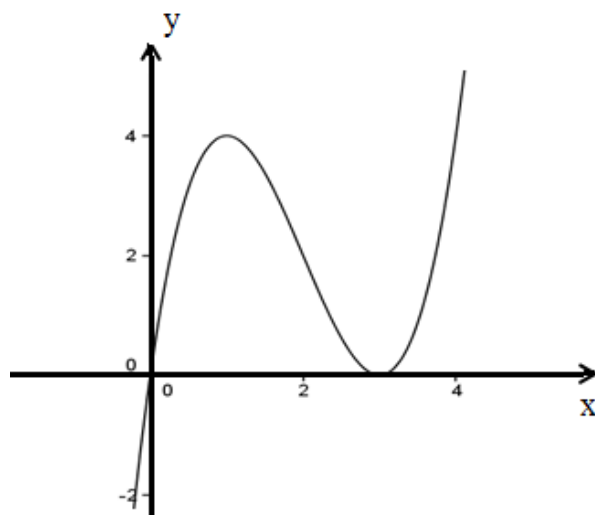
Section A: (10 marks)

Attempt Questions 1 – 10

Allow about 10 minutes for this section

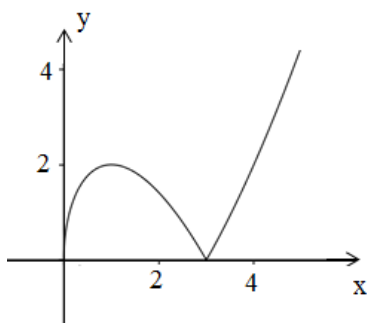
Use the multiple-choice answer sheet for questions 1 – 10 (detach from paper)

1. The graph $y = f(x)$ is shown below.

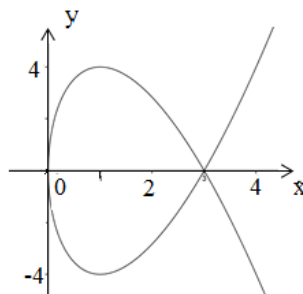


Which of the following graphs best represents $y^2 = f(x)$?

(A)

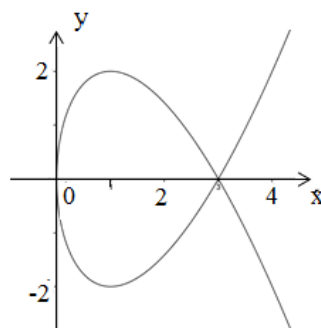
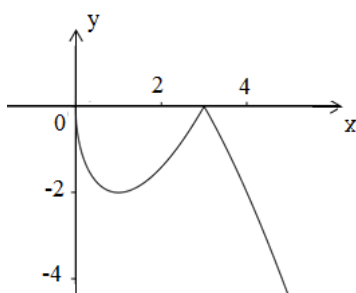


(B)



(D)

(C)



2. Let $z = a + ib$, where a and b are real and non-zero.

Which of the following statements is false?

- A. $z - \bar{z} = 2bi$
- B. $\arg(z) - \arg(\bar{z}) = 2\arg(z)$
- C. $|z| + |\bar{z}| = |z + \bar{z}|$
- D. $|z|^4 = |z|^2 |\bar{z}|^2$

3. If $f'(x) = 2\cos^2(x) - 1$ and $f\left(\frac{\pi}{6}\right) = 0$, then $f(x)$ is equal to

- A. $\frac{2\sin 2x - \sqrt{3}}{4}$
- B. $\frac{\sin 2x}{2}$
- C. $\frac{\sqrt{3}}{2} - \sin 2x$
- D. $\frac{\sin 2x}{2} + \frac{\sqrt{3}}{4}$

4. If $z = x + iy$, the locus of points that lie on a circle of radius 2 units centred at the origin on the Argand diagram can be represented by the equation,

- A. $z\bar{z} = 2$
- B. $(z + \bar{z})^2 - (z - \bar{z})^2 = 16$
- C. $\operatorname{Re}(z^2) + \operatorname{Im}(z^2) = 4$
- D. $\{\operatorname{Re}(z)\}^2 + \{\operatorname{Im}(z)\}^2 = 16$

5. What are the parametric equation of the ellipse $\frac{(x+1)^2}{4} + \frac{(y-3)^2}{2} = 1$?

- (A) $x = 1 - 2\cos\theta$, $y = 3 + \sqrt{2}\sin\theta$.
- (B) $x = -1 + 2\cos\theta$, $y = 2 + \sqrt{3}\sin\theta$.
- (C) $x = 2 + \sqrt{2}\cos\theta$, $y = 3 + \sqrt{2}\sin\theta$.
- (D) $x = -1 + 2\cos\theta$, $y = 3 + \sqrt{2}\sin\theta$.

6. z satisfies the equation $|z - \sqrt{2} - i\sqrt{2}| = 1$. Find the minimum value of $\arg(z)$.
- A. -75°
- B. -15°
- C. 15°
- D. 75°
7. The equation of the directrices of the hyperbola $4x^2 - 9y^2 - 8x - 18y - 41 = 0$ are:
- (A) $x = 1 \pm \frac{\sqrt{13}}{9}$
- (B) $x = 1 \pm \frac{9}{\sqrt{13}}$
- (C) $y = 1 \pm \frac{9}{13}$
- (D) $y = 1 \pm \frac{\sqrt{13}}{3}$
8. α, β and γ are the zeros of the polynomial $x^3 + 5x - 3$. Find the value of $\alpha(\alpha^2 - \beta\gamma) + \beta(\beta^2 - \gamma\alpha) + \gamma(\gamma^2 - \alpha\beta)$.
- A. -125
- B. 9
- C. 0
- D. 116
9. The function $f(x) = \frac{x^2}{x^2 - 4}$ has
- (A) A single vertical asymptote, two horizontal asymptotes and no turning points.
- (B) A single horizontal asymptote, two vertical asymptotes and no turning points.
- (C) A single vertical asymptote, two horizontal asymptotes and one turning point.
- (D) A single horizontal asymptote, two vertical asymptotes and one turning point.

10. The polynomial $P(z)$ has the equation $P(z) = z^4 - 4z^3 + Az + 20$, where A is a real number. Given that $3 + i$ is a zero of $P(z)$, which of the following expressions is $P(z)$ as a product of two real quadratic factors in real field.

A. $(z^2 - 2z + 2)(z^2 - 6z + 10)$

B. $(z^2 + 2z + 2)(z^2 + 6z + 10)$

C. $(z^2 - 2z + 2)(z^2 + 6z + 10)$

D. $(z^2 + 2z + 2)(z^2 - 6z + 10)$

Section B (90 marks)**Attempt Questions 11 – 16****Allow about 1 hours and 50 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 - 16, your responses should include relevant mathematical reasoning and/or calculations.

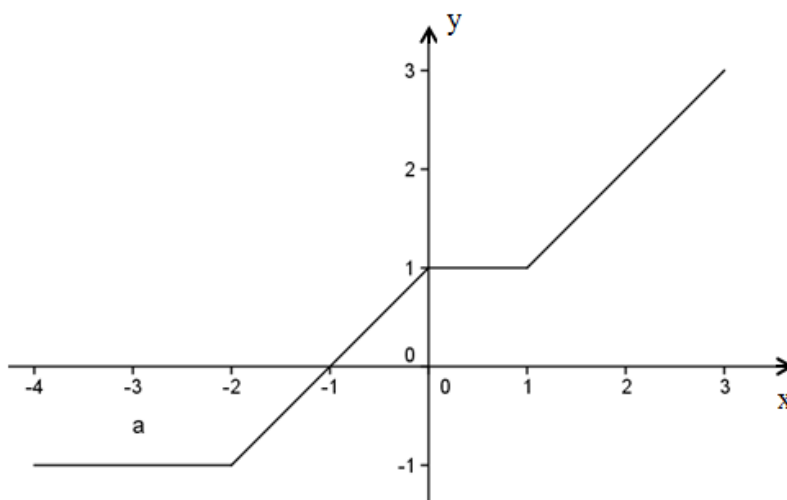
Question 11 (15 marks) Use a SEPARATE writing booklet

- a) Simplify $\left| \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} \right|$ 1
- b)
- i) Express $\frac{-1+i\sqrt{3}}{1-i}$ in modulus argument form 2
- ii) Hence find the exact value of $\tan \frac{11\pi}{12}$ 2
- c) Suppose $z = 1 + i\sqrt{3}$ and $w = z(\operatorname{cis} \alpha)$ where $-\pi \leq \alpha \leq \pi$. If w is purely imaginary and $\operatorname{Im}(w) > 0$, find the value of:
- i) α 2
- ii) $\arg(z + w)$ 2
- d)
- i) Find the square roots of $24 + 10i$. 1
- ii) Solve $z^2 + (1 + 3i)z - 8 - i = 0$ 2
- iii) Hence describe the locus of $|z - 2 + i| = |z^2 + (1 + 3i)z - 8 - i|$ 3

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

a)



The diagram is a sketch of the function $y = f(x)$. On separate diagrams sketch:

- | | | |
|------|-----------------------|----------|
| i) | $y = f(x) $ | 2 |
| ii) | $y = f(x)$ | 2 |
| iii) | $y = \frac{1}{f(x)}$ | 2 |
| iv) | $y = \sin^{-1}(f(x))$ | 2 |

The even function g is defined by

b)

$$g(x) = \begin{cases} 4e^{-x} - 6e^{-2x} & \text{for } x \geq 0 \\ g(-x) & \text{for } x \leq 0 \end{cases}$$

- | | | |
|------|---|----------|
| i) | Show that the graph of $g(x)$ has a maximum turning point at $\left(\ln 3, \frac{2}{3}\right)$. | 2 |
| ii) | Sketch the curve $y = g(x)$ and label the turning points, point of inflexion, asymptotes, and points of intersection with the axes. | 4 |
| iii) | Discuss the behaviour of the curve $y = g(x)$ at $x = 0$. | 1 |

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- a) α, β and γ are the roots of the equation $x^3 + 3px + q = 0$.
- i) Find a polynomial equation of degree 3 with roots $\frac{\alpha\beta}{\gamma}, \frac{\beta\gamma}{\alpha}$ and $\frac{\gamma\alpha}{\beta}$. 3
- ii) Hence find a relationship between p and q if $\alpha\beta = \gamma$. 1
- b) $(-1 + i\sqrt{3})$ is a zero of the polynomial $P(x) = x^4 + x^3 - 3x^2 - 14x - 20$.
Factorise $P(x)$ over the real field. 3
- c) The polynomial $P(x) = 16x^4 - 32x^3 + 16x^2 + kx - 5$, where k is an integer, has two rational roots which are equal but opposite signs and two non-real roots.
- i) If α is a non-real root of $P(x)$, show that $\operatorname{Re}(\alpha) = 1$ and $|\alpha| > 1$. 2
- ii) If the rational roots are $\pm \beta$, deduce that $\beta^2 < \frac{5}{16}$. 2
- iii) Find the rational roots and the value of k . 2
- iv) Factorise $P(x)$ into irreducible factors with integer coefficients. 2

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- a. Consider the conic defined by the equation $\frac{x^2}{19-l} + \frac{y^2}{7-l} = 1$
- i) Determine the real values of l for which the equation defines an ellipse and a hyperbola. 2
 - ii) Sketch the curve corresponding to the value $l = 3$. 2
 - iii) Describe how the shape of this curve changes as l increases from 3 towards 7. What is the limiting position of the curve as l is approached? 2
- b. T is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre O.
- A line drawn through O, parallel to the tangent to the ellipse at T, meets the ellipse at M and N. Find:
- i) the equation of MN. 2
 - ii) the perpendicular distance from T to MN. 4
 - iii) Hence prove that the area of the triangle TMN is independent of the position of T 3
- a

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet

- a) (i) Use De Moivre's theorem to express $\sin 3\theta$ in terms of $\sin \theta$. 2
- (ii) Use this result to solve $8x^3 - 6x + \sqrt{3} = 0$. 2
- (iii) Hence deduce that $\sin \frac{\pi}{9} + \sin \frac{7\pi}{9} = \sin \frac{5\pi}{9}$. 2

- b) z_1 and z_2 are the zeros of the equation $z^2 - (1 + i)z + (1 + i\sqrt{3}) = 0$.
- Find the value of : (i) $\arg z_1 + \arg z_2$ 1
- (ii) $|z_1 + z_2|$ 1

- c) Given that $z = \cos \theta + i \sin \theta$, show that $\cos^4 \theta + \sin^4 \theta = \frac{1}{4} (\cos 4\theta + 3)$ 2

- d) Given that $P \equiv e^{i\theta} \equiv \cos \theta + i \sin \theta$ where $-\pi \leq \theta \leq \pi$,
- (i) Evaluate $\sqrt[4]{-1}$ correct to 4 decimal places. 2
- (ii) By considering the sum of the series $1 + P + P^2 + P^3 + \dots$ to n terms, show that: 3

$$1 + \cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos(n-1)\theta =$$

$$\frac{[(1 - \cos \theta)(1 - \cos(n\theta)) \sin \theta \sin(n\theta)]}{2 - 2\cos \theta}$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet

a) i) Show that $\sin x + \sin 3x = 2\sin 2x \cos x$. 2

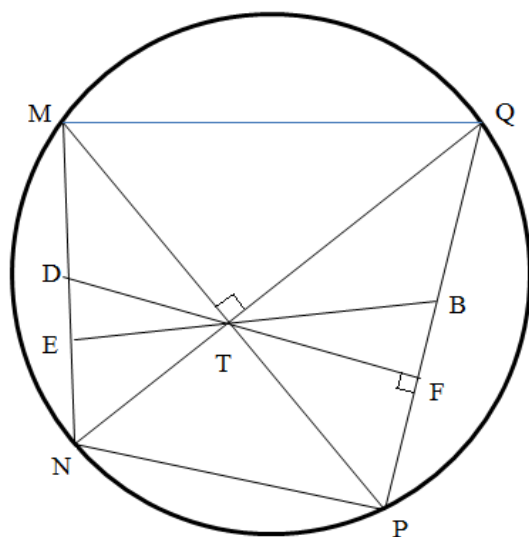
ii) Hence otherwise find all the solutions of $\sin x + \sin 2x + \sin 3x = 0$ for $0 \leq x \leq 2\pi$. 2

b) Given $f(x) = x - \log_e (1 + x^2)$

i) Show that $f'(x) \geq 0$ for all values of x . 2

ii) Hence deduce that $e^x > 1 + x^2$ for all positive values of x . 2

c)



In the diagram above, the diagonals MP and NQ of the cyclic quadrilateral MNPQ intersect at a right angle at the point T. The perpendicular TF to QP meets MN at D. The median from T to the midpoint of QP meets MN at E,

i) Show that the point D is the midpoint of MN. 3

ii) Show that $\angle TEN = 90^\circ$. 2

iii) Let A and C be the midpoints of NP and MQ respectively. Show that the quadrilateral ABCD is a rectangle. 2

END OF EXAMINATION

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Student Number

Mathematics

Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 10. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct

- Start Here** →
1. A ☐ B ☐ C ☐ D ☐
 2. A ☐ B ☐ C ☐ D ☐
 3. A ☐ B ☐ C ☐ D ☐
 4. A ☐ B ☐ C ☐ D ☐
 5. A ☐ B ☐ C ☐ D ☐
 6. A ☐ B ☐ C ☐ D ☐
 7. A ☐ B ☐ C ☐ D ☐
 8. A ☐ B ☐ C ☐ D ☐
 9. A ☐ B ☐ C ☐ D ☐
 10. A ☐ B ☐ C ☐ D ☐

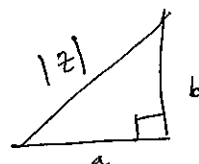
Q11 (a) $\left| \cos \frac{2\pi}{5} + j \sin \frac{2\pi}{5} \right|$

$$= \sqrt{\cos^2 \frac{2\pi}{5} + \sin^2 \frac{2\pi}{5}}$$

$$= \sqrt{1}$$

$$= \underline{\underline{1}}$$

$z = a + bi$
 $|z| = \sqrt{a^2 + b^2}$



correct answer

→ 1 mark

(b) (i) $\frac{-1 + j\sqrt{3}}{1 - j}$

Mod arg.
 Mod - arg.

Many people opting for the long process.

$$= \frac{2 \cos(\frac{2\pi}{3})}{\sqrt{2} \cos(-\frac{\pi}{4})}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \cos(\alpha - \beta)$$

1 mark for expressing both in mod-arg form

$$= \sqrt{2} \cos(\frac{2\pi}{3} + \frac{\pi}{4})$$

$$= \sqrt{2} \cos(\frac{11\pi}{12})$$

1 mark for correct answer

(ii) $\frac{-1 + j\sqrt{3}}{1 - j} = \frac{(-1 + j\sqrt{3})(1 + j)}{2}$

$$= \left(\frac{\sqrt{3}-1}{2} \right) + j \left(\frac{\sqrt{3}-1}{2} \right)$$

1 mark for $a + jb$ form

$$\Rightarrow \sqrt{2} \cos(\frac{11\pi}{12}) = \left(\frac{\sqrt{3}-1}{2} \right) + j \left(\frac{\sqrt{3}-1}{2} \right)$$

Equating the imaginary parts

$$\sqrt{2} \sin(\frac{11\pi}{12}) = \frac{\sqrt{3}-1}{2} \quad \text{--- (1)}$$

Equating the real parts

$$\sqrt{2} \cos(\frac{11\pi}{12}) = -\frac{\sqrt{3}-1}{2} \quad \text{--- (2)}$$

1 mark for working & correct answer

(1) ÷ (2)

$$\tan(\frac{11\pi}{12}) = - \frac{(\sqrt{3}-1)}{(\sqrt{3}+1)}$$

$$= \underline{\underline{\sqrt{3}-2}}$$

Q4

$$(c) \quad Z = 1 + j\sqrt{3} \\ = 2 \cos \pi/3$$

$$W = Z \cos \alpha = 2 \cos (\pi/3 + \alpha)$$

(i) W is purely imaginary

$$\Rightarrow \operatorname{Re}(W) = 0$$

$$\Rightarrow 2 \cos (\pi/3 + \alpha) = 0$$

$$\left. \begin{aligned} \pi/3 + \alpha &= \pi/2 \\ \alpha &= \pi/6 \end{aligned} \right\}$$

1 mark for correctly equating

1 mark for correct answer

$$(ii) \quad Z = 1 + j\sqrt{3}, \quad W = 2j$$

$$\therefore \cancel{Z+W} = \cancel{1+2j}$$

$$\therefore Z+W = 1 + (\sqrt{3}+2)j$$

$$\therefore \arg(Z+W) = \tan^{-1}(2+\sqrt{3})$$

$$\approx 1.31 \text{ radians}$$

$$= \frac{5\pi}{12}$$

1 mark for evaluating $(Z+W)$ correctly

1 mark for correct answer

Q11 (d)

$$(i) \sqrt{24+10i} = \sqrt{(5+i)^2} \\ = \underline{\underline{\pm (5+i)}}$$

→ 1 mark for correct working + answer

Must learn an easier method to do this question.

$$(ii) z^2 + (1+3i)z - 8-i = 0$$

$$z = \frac{-(1+3i) \pm \sqrt{(1+3i)^2 + 4(8+i)}}{2}$$

$$= \frac{-(1+3i) \pm \sqrt{24+10i}}{2}$$

$$= \frac{-(1+3i) \pm (5+i)}{2}$$

$$= \underline{\underline{(2-i) \text{ or } (-3-2i)}}$$

→ 1 mark for using quadratic formula correctly

→ 1 mark for correct answer

(iii)

$$|z-2+i| = |z^2 + (1+3i)z - 8-i|$$

$$= |(z-2+i)(z+3+2i)|$$

→ 1 mark for identifying this

→ 1 mark

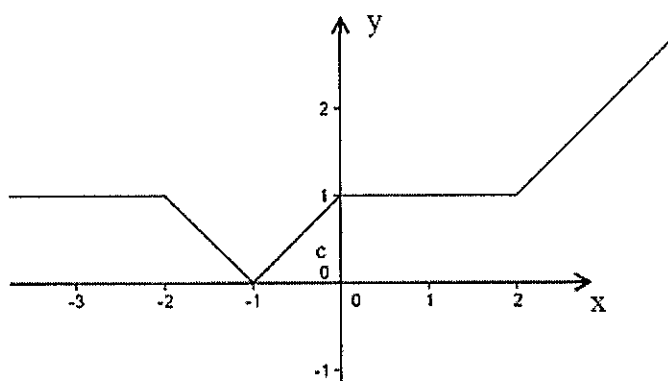
⇒ $|z+3+2i| = 1$
 ⇒ locus of z is a circle on an Argand diagram with centre at $(-3, -2)$ and radius 1 unit.

→ 1 mark for correctly identifying.

Many students failed to see make the connection between Q(ii) and (iii)

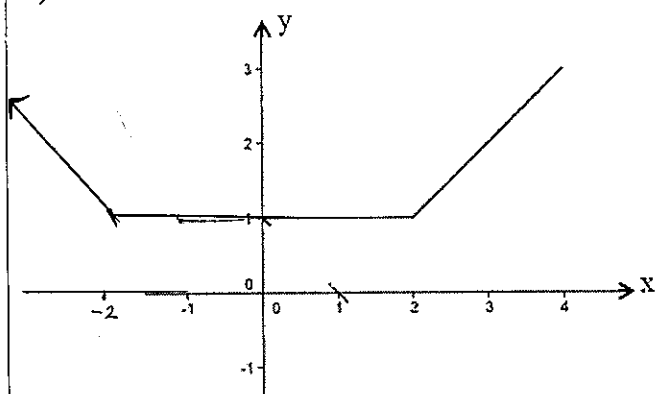
Q12(a)

i)



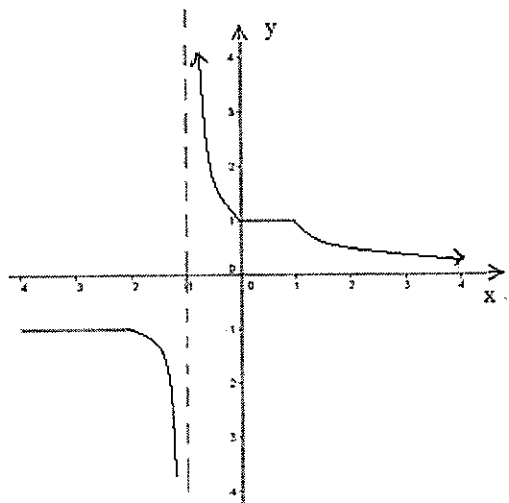
2 - each correct graph with correct label of values

ii)



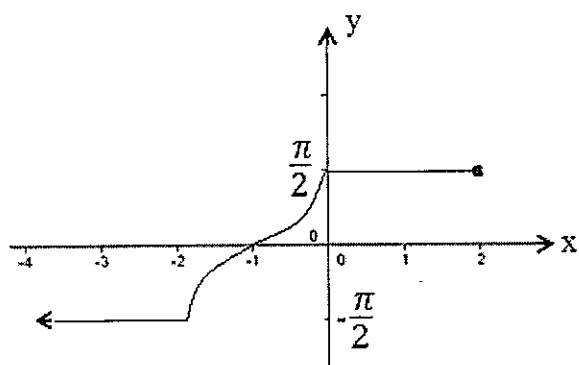
- Discards the ~~left~~ $x < 0$
- reflects to right

iii)



- $x > 0$ Correct shape.
- $x < 0$ Correct shape.

iv)



- Capped between $y = -1$ to 1 .
- Correct slope 1.

The shape of ~~Start~~ $f(x)$ between $-2 \leq x \leq 1$ was a problem

Many students forgot that the original graph was nothing but $y = x$ in this region and hence $f(x)$ should be the shape of the original

(b)

$$g'(x) = \frac{12}{e^{2x}} - \frac{4}{e^x}$$

$$g''(x) = \frac{4}{e^x} - \frac{24}{e^{2x}}$$

For stationary point $g'(x) = 0$

$$\frac{12}{e^{2x}} - \frac{4}{e^x} = 0$$

$$12 - 4e^x = 0$$

$$e^x = 3$$

$$x = \ln 3$$

$$y = 4e^{-\ln 3} - 6e^{-2\ln 3}$$

$$= \frac{4}{3} - \frac{6}{9} = \frac{2}{3}$$

$$g''(\ln 3) = \frac{4}{3} - \frac{24}{9}$$
$$= -\frac{12}{9} < 0$$

$\therefore (\ln 3, \frac{2}{3})$ is a maximum point.

ii) For point of inflection $g''(x) = 0$

$$4e^{-x}(1 - 6e^{-x}) = 0$$

$$4e^{-x} = 24$$

$$e^x = 6$$

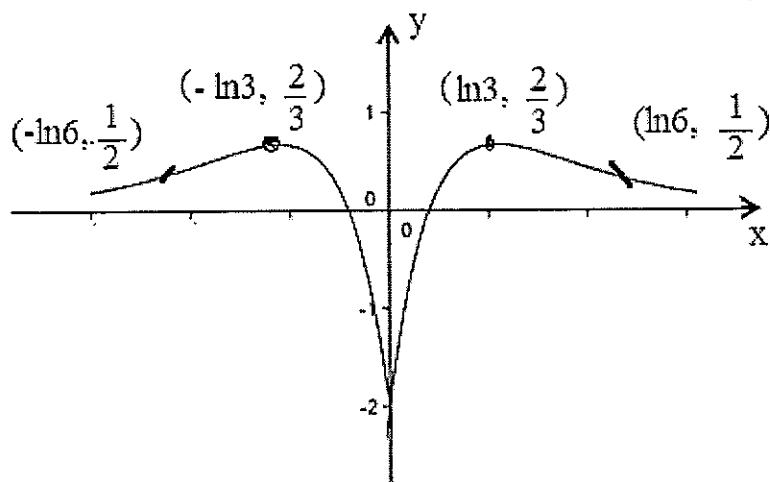
$$x = \ln 6$$

$$y = 4e^{-\ln 6} - 6e^{-2\ln 6}$$

$$= \frac{4}{6} - \frac{6}{36} = \frac{1}{2}$$

x	$\ln 5$	$\ln 6$	$\ln 7$
$g''(x)$	$-\frac{8}{5}$	0	0.004

$\therefore (\ln 6, \frac{1}{2})$ is the point of inflexion.



iii) At $x = 0$, $g(x) = -2$, $g'(x) = 8$ and $g''(x) = -8$
The curve has a cusp at $(0, -2)$

1- correctly
differentiate $g'(x)$

1-correct answer

2-correct answer
with all working

2-correct graphs and
labelling

Must note that this
is an even function. so doing
 $x > 0$ is enough and for $x < 0$
the graph can be reflected

Must say why the function is
undifferentiable at $x = 0$.

Q13 (a)

$$x^3 + 3px + q = 0$$

roots α, β, γ

$$(i) \alpha\beta\gamma = -q \Rightarrow \frac{\alpha\beta}{\gamma} = -\frac{q}{\gamma^2}$$

$$\therefore \frac{\alpha\beta}{\gamma}, \frac{\beta\gamma}{\alpha} \text{ and } \frac{\gamma\alpha}{\beta} \text{ are roots}$$

$$\Rightarrow -\frac{q}{\alpha^2}, -\frac{q}{\beta^2} \text{ and } -\frac{q}{\gamma^2} \text{ are the roots}$$

$$-\frac{q}{\alpha^2} = \alpha$$

$$\therefore \alpha = \sqrt{-\frac{q}{\alpha}}$$

$$p(\gamma\alpha) = 0$$

$$\Rightarrow p\left(\sqrt{-\frac{q}{\alpha}}\right) = 0$$

$$\Rightarrow \left(\sqrt{-\frac{q}{\alpha}}\right)^3 + 3p\sqrt{-\frac{q}{\alpha}} + q = 0$$

$$\sqrt{-\frac{q}{\alpha}} \left[-\frac{q}{\alpha} + 3p\right] = -q$$

squaring both sides,

$$-\frac{q}{\alpha} \left[-\frac{q}{\alpha} + 3p\right]^2 = q^2$$

$$\Rightarrow \frac{q^2}{\alpha^2} - \frac{6pq}{\alpha} + 9p^2 = -q\alpha$$

$$\Rightarrow q\alpha^3 + 9p^2\alpha^2 - 6pq\alpha + q^2 = 0$$

$$(ii) \alpha\beta = \gamma \Rightarrow \frac{\alpha\beta}{\gamma} = 1$$

$$\Rightarrow q + 9p^2 - 6pq + q^2 = 1$$

$$\Rightarrow (3p - q)^2 = -q$$

1 mark
getting the
root in this
form

This is a very
routine question;
but some students
have not bothered
to learn it.

1 mark
for correct
substitution.

In general done well.

1 mark
for correct
working and
correct eqⁿ

In fact, substitution is
the easiest way to eliminate
 α .

1 mark
for correct
relationship

Many students
failed to see
that $\alpha = 1$ is
a root, $\therefore$ satisfied
the equation.

Q13

(b) $P(x) = x^4 + x^3 - 3x^2 - 14x - 20$
 Since the coefficients of $P(x)$
 are all real,

$(-1 + i\sqrt{3})$ is a zero $\Rightarrow$ its complex conjugate
 $(-1 - i\sqrt{3})$ is also a zero.

$\Rightarrow [x - (-1 + i\sqrt{3})][x - (-1 - i\sqrt{3})]$ is a factor
 of $P(x)$

$\Rightarrow (x^2 + 2x + 4)$ is a factor of $P(x)$.

$$x^4 + x^3 - 3x^2 - 14x - 20$$

$$= (x^2 + 2x + 4)(x^2 - x - 5)$$

$$= (x^2 + 2x + 4) \left[x - \frac{1 - \sqrt{21}}{2} \right] \left[x - \frac{1 + \sqrt{21}}{2} \right]$$

Many students still
 have problem to write
 the quadratic with given
 roots
 $\rightarrow$ 1 mark for
 using them

Many choose to guess it,
 rather than working it out.

1 mark for
 arriving at this

1 mark for
 correct answer

Q13 (c)

~~(b)~~ $P(x) = 16x^4 - 32x^3 + 16x^2 + kx - 5$

Since the coefficients are all
(i) real,

α is a non-real root $\Rightarrow$ its conjugate
 $\bar{\alpha}$ is also a root

$\therefore$ let the roots be

$\beta, -\beta, \alpha$ and $\bar{\alpha}$

Sum of roots $= -\frac{(-32)}{16}$

$\Rightarrow \beta + -\beta + \alpha + \bar{\alpha} = 2$

$\Rightarrow 2 \operatorname{Re}(\alpha) = 2$

$\Rightarrow \underline{\underline{\operatorname{Re}(\alpha) = 1}}$

$\Rightarrow |\alpha| = \sqrt{1 + [\operatorname{Im}(\alpha)]^2}$
 $\underline{\underline{> 1}}$

1 mark
correct working

Generally
well done.

1 mark
for this
reasoning

Those students
who wrote
 $\alpha = x + iy$ got
this question right.

(ii) Products of roots $= -\frac{5}{16}$

$\Rightarrow \beta \cdot -\beta \cdot \alpha \cdot \bar{\alpha} = -\frac{5}{16}$

$\Rightarrow -\beta^2 \cdot |\alpha|^2 = -\frac{5}{16}$

$\Rightarrow \beta^2 = \frac{5}{16|\alpha|^2}$

$\underline{\underline{< \frac{5}{16} \left(\because \frac{1}{|\alpha|^2} < 1 \right)}}$

1 mark for
this working

well done!

1 mark
for correct
reasoning

This is a
"Show that"
question. Every
line of logical
progression must
be shown.

Q13

(iii) $\beta, -\beta$ are roots

$$\therefore P(\beta) = 0, P(-\beta) = 0$$

$$\Rightarrow P(\beta) + P(-\beta) = 0$$

$$\Rightarrow 16\beta^4 + 16\beta^2 - 5 = 0$$

$$\Rightarrow (4\beta^2 - 1)(4\beta^2 + 5) = 0$$

Since $\beta \neq 0$ and $4\beta^2 + 5 \neq 0$

$$\Rightarrow 4\beta^2 = 1$$

$$\Rightarrow \beta = \frac{1}{2} \text{ or } \beta = -\frac{1}{2}$$

$\therefore$ The unknown roots are $\frac{1}{2}$ and $-\frac{1}{2}$.

$$P\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow 16\left(\frac{1}{2}\right)^4 - 32\left(\frac{1}{2}\right)^3 + 16\left(\frac{1}{2}\right)^2 + k\left(\frac{1}{2}\right) - 5 = 0$$

$$\Rightarrow \underline{\underline{k = 8}}$$

Hardly anybody could see this logic.

1 mark working & correct answer

Very poorly done.

In general,

Q (iii) & (iv)

were N/A.

1 mark correct answer

$$(iv) P(x) = 16x^4 - 32x^3 + 16x^2 + 8x - 5$$

$$= (2x - 1)(2x + 1)(4x^2 + 8x + 5)$$

$$= (2x - 1)(2x + 1)(2x - 2 - i)(2x - 2 + i)$$

1 mark

1 mark.

EX2 – Half Yearly – 2015

Section A

1. D 2. C 3. A 4. B 5. D
6. C 7. B 8. C 9. D 10. D

Section B

Q14(a)

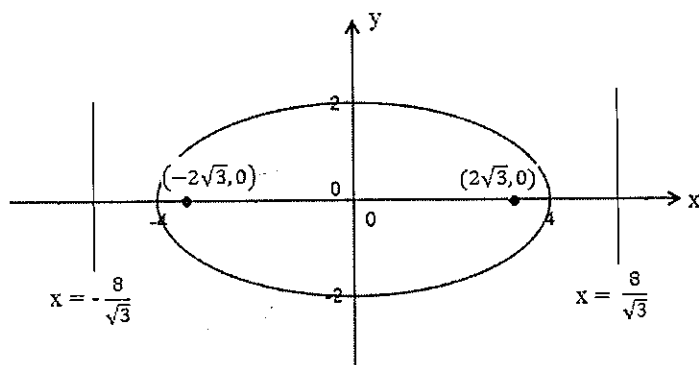
- (i) For the ellipse
 $19 - l > 0$ and $7 - l > 0$
 $l < 19$ and $l < 7$

$$\therefore l < 7$$

For the hyperbola

- $19 - l > 0$ and $7 - l < 0$
 $\therefore 7 < l < 19$

(ii)



- (iii) As l increases from 3 towards 7 the shape of the curve approaches an interval where
 $-2\sqrt{3} \leq x \leq 2\sqrt{3}$

As l approaches 7

$$a^2 \Rightarrow 19 - 7, \quad a \Rightarrow 2\sqrt{3}$$

$$\text{and} \quad b^2 \Rightarrow 7 - 7 \Rightarrow 0$$

$\therefore$ The limiting position of the curve is SS' of the above ellipse.

(b)

$$\frac{dx}{d\theta} = -a \sin \theta \quad \frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{b \cos \theta}{-a \sin \theta}$$

MN // tangent at T

1 - correct answer

1 - correct answer

2 - correct graphs and label of value

1 - correct answer

1 - correct answer

1 - correctly differentiate & find m

Using $0 < e < 1$ for ellipse and $e > 1$ for hyperbola and trying to solve for l makes this question, dense in working.

If you said y - you got 1 mark, but needed to say the limiting value: x to get the full marks.

Many students wasted time trying to find the equation of the tangent. But it has nothing to do with this question.

$\therefore m_{\text{of MN}} = -\frac{b \cos \theta}{a \sin \theta}$ The equation of MN is: $y = -\frac{b \cos \theta}{a \sin \theta} x$ (1) ii) Sub (1) into $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\Rightarrow \frac{x^2}{a^2} + \frac{x^2 b^2 \cos^2 \theta}{b^2 a^2 \sin^2 \theta} = 1$ $\Rightarrow \frac{x^2 (\sin^2 \theta + \cos^2 \theta)}{a^2 \sin^2 \theta} = 1$ $\therefore$ The coordinates of MN are $x = \pm a \sin \theta$ $y = \frac{-b \cos \theta}{a \sin \theta} \times a \sin \theta$ $y = \pm b \cos \theta$ The perpendicular distance TP to MN at P is : $\left \frac{\frac{b \cos \theta \times a \cos \theta}{a \sin \theta} + b \sin \theta}{\sqrt{1 + \frac{b^2 \cos^2 \theta}{a^2 \sin^2 \theta}}} \right = \left \frac{ab \cos^2 \theta + ab \sin^2 \theta}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}} \right $ $= \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$ iii) The distance MN is $\sqrt{(2a \sin \theta)^2 + (2b \cos \theta)^2} = \sqrt{4(a^2 \sin^2 \theta + b^2 \cos^2 \theta)}$ $= 2\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$ Area of ΔTMN is $= \frac{1}{2} \times 2\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} \times \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$ $= ab \text{ which is independent of the position of T}$	1- correct answer 1- for each correct coordinates of M & N 1-correct formula & substitution of perpendicular distance 1-correct answer of TP 1- correct answer of MN 1 - Finding area of triangle 1 - correct answer & conclusion	Also, if it is passing through the origin, its' general form is $y=mx$. No need to use point-gradient formula and make it complicated. Again, those who used parametric form, got it much faster.
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Q15 (a)

(i) $(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$

(ii) $(\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i(3 \cos^2 \theta \sin \theta - \sin^3 \theta) = \cos 3\theta + i \sin 3\theta$

1 mark for correct expansion & using De Moivre's theorem

Matching the imaginary parts

$$\begin{aligned} \sin 3\theta &= 3 \cos^2 \theta \sin \theta - \sin^3 \theta \\ &= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta \\ &= 3 \sin \theta - 4 \sin^3 \theta \end{aligned}$$

1 mark for equating real parts & simplifying

(ii) $8x^3 - 6x + \sqrt{3} = 0$

let $x = \sin \theta$

$$\begin{aligned} 8 \sin^3 \theta - 6 \sin \theta &= -\sqrt{3} \\ \Rightarrow 2 \sin 3\theta &= \sqrt{3} \\ \sin 3\theta &= \frac{\sqrt{3}}{2} \end{aligned}$$

1 mark using the 1st part to get them

$$\Rightarrow \theta = \frac{n\pi}{3} + (-1)^n \frac{\pi}{6}$$

$$\begin{aligned} n=0 \quad \theta &= \frac{\pi}{6} \Rightarrow x = \sin \frac{\pi}{6} \\ n=-1 \quad \theta &= -\frac{\pi}{3} - \frac{\pi}{6} \Rightarrow x = \sin(-\frac{\pi}{2}) = -1 \\ n=1 \quad \theta &= \frac{\pi}{3} - \frac{\pi}{6} \Rightarrow x = \sin \frac{\pi}{6} = \frac{1}{2} \end{aligned}$$

1 mark

$\therefore$ The roots are $x = \sin \frac{\pi}{6}, -1, \sin \frac{\pi}{6}$

(iii) Sum of roots = 0

$$\begin{aligned} \Rightarrow \sin \frac{\pi}{6} - 1 + \sin \frac{\pi}{6} &= 0 \\ \Rightarrow \sin \frac{\pi}{6} + \sin \frac{\pi}{6} &= 1 \end{aligned}$$

1 mark

A large number of students stopped with $n=0, 1, 2$ yielding $\frac{2\pi}{3}, \frac{7\pi}{6}$. But they did not realise $\sin \frac{2\pi}{3} = \sin \frac{7\pi}{6}$. I think, Q(iii) should give you the clue.

Q15 (b)

$$z^2 - (1+i)z + (1+i\sqrt{3}) = 0$$

(i) Product of roots $= 1+i\sqrt{3}$

$$\Rightarrow z_1 z_2 = 1+i\sqrt{3}$$

$$\Rightarrow \arg(z_1 z_2) = \arg(1+i\sqrt{3})$$
$$= \frac{\pi}{63}$$

$$\Rightarrow \arg(z_1) + \arg(z_2) = \frac{\pi}{63}$$

1 mark
correct
answer

1 mark
correct
answer

(ii) Sum of roots $= 1+i$

$$\Rightarrow z_1 + z_2 = 1+i$$

$$\Rightarrow |z_1 + z_2| = |1+i|$$
$$= \underline{\underline{\sqrt{2}}}$$

1 mark
correct
answer

(d)

(i) $\cos \theta + i \sin \theta = e^{i\theta}$

when $\theta = \pi$

$$-1 = e^{i\pi}$$

$$\Rightarrow \sqrt[4]{-1} = \sqrt[4]{e^{i\pi}}$$
$$= e^{i\pi/4}$$

$$= \underline{\underline{23.1407}}$$

1 mark for
correct steps
to get this

1 mark
correct
answer

15.
16(c) $z = \cos \theta + i \sin \theta$

$$z^n = \cos(n\theta) + i \sin(n\theta) \quad \text{--- (1)}$$

$$\frac{1}{z^n} = \cos(n\theta) - i \sin(n\theta) \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow z^n + \frac{1}{z^n} = 2 \cos(n\theta)$$

$$\textcircled{1} - \textcircled{2} \Rightarrow z^n - \frac{1}{z^n} = 2i \sin(n\theta)$$

$$\therefore 16 \cos^4 \theta + 16 \sin^4 \theta$$

$$= \left(z + \frac{1}{z}\right)^4 + \left(z - \frac{1}{z}\right)^4$$

$$= 2 \left[z^4 + 6 + \frac{1}{z^4} \right]$$

$$= 2 [2 \cos 4\theta + 6]$$

$$\therefore \cos^4 \theta + \sin^4 \theta = \frac{1}{4} [\cos 4\theta + 3]$$

1 mark for getting this

1 mark for correct simplifying to get the answer

When you need to find powers of $\cos \theta$ and $\sin \theta$, you use the ~~for~~ result

$$z + z^{-1} = 2 \cos \theta$$

$$z - z^{-1} = 2i \sin \theta$$

Then, break it down to

$$z^n + z^{-n} = 2 \cos n\theta$$

$$z^n - z^{-n} = 2i \sin n\theta$$

Q15 (ii)

$$1 + p + p^2 + p^3 + \dots + p^{n-1} = \frac{1 - p^n}{1 - p}$$

~~Sum of GP with 1st~~
Sum of the first n terms of a GP
with first term 1 and common ratio p

Matching the real parts

$$1 + \cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos(n-1)\theta = \operatorname{Re} \left[\frac{1 - p^n}{1 - p} \right]$$

1 mark
Using sum
of series to
get term

$$\frac{1 - p^n}{1 - p} = \frac{(1 - \cos n\theta) - i \sin n\theta}{(1 - \cos \theta) - i \sin \theta} \times \frac{(1 - \cos \theta) + i \sin \theta}{(1 - \cos \theta) + i \sin \theta}$$

1 mark
1 mark

$$= \frac{[(1 - \cos \theta)(1 - \cos n\theta) + \sin \theta \sin n\theta] + i [\dots]}{(1 - \cos \theta)^2 + \sin^2 \theta}$$

~~1 mark~~

$$\Rightarrow \operatorname{Re} \left(\frac{1 - p^n}{1 - p} \right) = \frac{(1 - \cos \theta)(1 - \cos n\theta) + \sin \theta \sin n\theta}{(1 - \cos \theta)^2 + \sin^2 \theta} = \frac{(1 - \cos \theta)(1 - \cos n\theta) + \sin \theta \sin n\theta}{2 - 2 \cos \theta}$$

1 mark
for
correct
working

Q16(a)

(i)
$$\begin{aligned} L.H.S &= \sin(2x+x) + \sin(2x-x) \\ &= \sin 2x \cos x + \cos 2x \sin x \\ &\quad + \sin 2x \cos x - \cos 2x \sin x \\ &= 2\sin 2x \sin x \\ &= R.H.S \end{aligned}$$

(ii)
$$\begin{aligned} \sin x + \sin 2x + \sin 3x &= 0 \quad \text{Sub (i)} \\ 2\sin 2x \cos x + \sin 2x &= 0 \\ \sin 2x(2\cos x + 1) &= 0 \end{aligned}$$

$$\begin{aligned} \sin 2x &= 0 \\ 2x &= 0, \pi, 2\pi, 3\pi, 4\pi \\ x &= 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi \end{aligned}$$

$$\begin{aligned} \text{and } 2\cos x &= -1 \\ x &= \frac{2\pi}{3}, \frac{4\pi}{3} \end{aligned}$$

(b)
(i)

$$\begin{aligned} f'(x) &= 1 - \frac{2x}{1+x^2} \\ &= \frac{1+x^2-2x}{1+x^2} \\ &= \frac{(1-x)^2}{1+x^2} \end{aligned}$$

$$(1+x)^2 \text{ and } 1+x^2 > 0 \text{ for all } x$$

$\therefore f(x)$ is monotonic increasing.

(ii)

$$\begin{aligned} \text{When } x=0, f(x) &= -\log_e 1 \\ &= 0 \end{aligned}$$

$$\therefore f(x) > 0 \text{ for all } x > 0$$

Hence

$$\begin{aligned} x - \log_e(1+x^2) &> 0 && \text{for all } x > 0 \\ \log_e(1+x^2) &< x && \text{for all } x > 0 \end{aligned}$$

$$\therefore e^x > 1+x^2$$

1 - correct method

1 - correctly show
L.H.S = R.H.S

1 - Showing correct
substitution

1 - correct answer

1 - correctly
differentiation

1 - correct
explanation

1 - finding the value
of $f(x)$ when $x=0$

1 - correct reasoning

Must learn the
"Sums as products"
formulae. These are
very useful for
simplifying expressions

$$\sin 2x = 0$$

$$0 \leq x \leq 2\pi$$

$$\therefore 0 \leq 2x \leq 4\pi$$

Many students
lost the solutions

$$\text{at } \frac{3\pi}{2} \text{ and } 2\pi$$

as they mistook

$$0 \leq 2x \leq 2\pi$$

$$f'(x) > 0$$

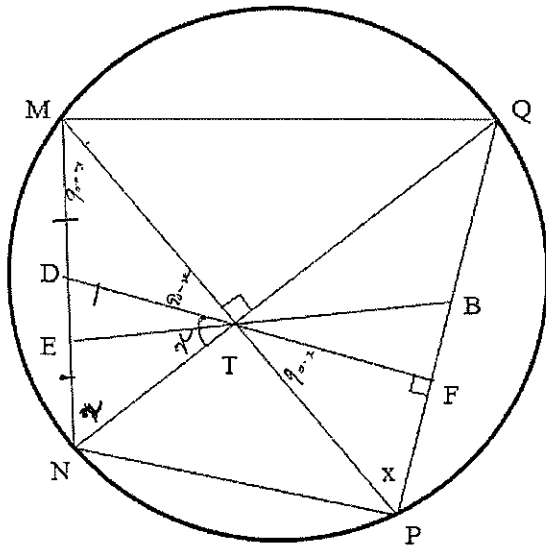
does not mean

$$f(x) > 0. \text{ of course,}$$

it is an
increasing function.

You must find
the minimum
value of this
function and show
that it is > 0 .

(b)



(i)

Let $\angle MPQ = x$

$$\angle \text{MNQ} = \angle \text{MPQ} \quad (\angle \text{s in the same segment}) \quad (1)$$

$$= x$$

$$\begin{aligned}\angle \text{PTF} &= 90^\circ - x && (\angle \text{sum in right angled triangle}) \\ &= \angle \text{MTD} && (\text{vertically opposite angles})\end{aligned}$$

$$\begin{aligned}\angle DMT &= 90^0 - \angle MNQ \quad (\angle \text{sum in right angled triangle}) \\ &= 90^0 - x \\ &= \angle MTD \quad (2)\end{aligned}$$

$\therefore \Delta DMT$ is isosceles (2 equal angles)

Hence $DT = DM$ (3)

$$\begin{aligned}\angle DTM &= 90^\circ - \angle MTD = 90^\circ - (90^\circ - x) = x \\ &= 90^\circ - x \quad \text{from (2)} \\ &= \angle DNT \quad \text{from (1)}\end{aligned}$$

$\therefore \triangle DNT$ is isosceles (2 equal angles)

Hence $DT = DN$

$\therefore DM = DN$, hence D is the mid-point of MN

(ii)

$\angle QTP = 90^\circ$ and B is the mid-point of PQ

$\therefore \Delta TQP$ is in a semicircle with TB, BQ and BP are radii.
Hence ΔQBT is isosceles.

$$\begin{aligned}\therefore \angle QTB &= (\text{vertically opposite angles}) \angle TQP \\ &= 90^\circ - x \quad (\angle \text{sum in } \triangle TPQ) \\ &= \angle ETN \quad (\text{vertically opposite angles})\end{aligned}$$

1- first three steps shown

1- Proving the triangle is isosceles

1- showing the result

1

$\angle TEN = 180^\circ - x - (90^\circ - x)$ ($\angle$ sum in a triangle) $= 90^\circ$ (iii) In $\triangle DMC$ and $\triangle MNQ$ $\angle M$ is common $\frac{MD}{MN} = \frac{MC}{MQ} = \frac{1}{2}$ (D & C are mid-points of MN & MQ) $\therefore \triangle DMC \sim \triangle MNQ$ (two pairs of corresponding lengths are in same ratio and one pair of equal corresponding angles) Hence corresponding angles are equal $\therefore \angle MCD = \angle MQN$ $\therefore DC \parallel NQ$ (corresponding angles are equal on // lines) Similarly $AB \parallel NQ$, $BC \parallel MP$ and $AD \parallel MP$ $\therefore DC \parallel AB$ and $AD \parallel BC$ Hence ABCD is a parallelogram (opposite sides are parallel) $MP \perp NQ$ (Given) But $BC \parallel MP$ and $DC \parallel NQ$ $\therefore DC \perp BC$ Hence ABCD is a rectangle (the parallelogram ABCD has one right angle).	1-showing the result	
	1-showing the quad. Is a parallelogram	
	1-showing the parallelogram is a rectangle	